Shanmukh Upadhyayula

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Education

University of Illinois Urbana-Champaign GPA: 3.95/4.0

Master of Computer Science (MCS), Planned *August 2027 – May 2028*

Bachelor of Science in Computer Science & Geographic Information Science *August 2024 – May 2027*

- Involvement: ACM SIG Project Lead for HCI and AI, Virtual Reality Physics Simulation Developer
- Relevant Coursework: Data Structures, Algorithms & Models of Computation, Computer Systems, Database Management in the Cloud, Applied Linear Algebra, Spatial Analysis, Human-LLM Interaction

Experience

UIUC CyberGIS Lab Urbana, IL

Machine Learning Engineering Intern *September 2026 – Present*

- Benchmarking the lab's LLM-based geospatial AI agent on all 50 GeoAnalystBench tasks, showing it matches published LLMs on code but over-plans on 43 of 48 tasks
- Uncovering 4 bugs that break core features of the live agent by user-testing dataset search, map projections, raster data, file uploads, and adversarial prompts

Semantic Segmentation Failure Analysis Urbana, IL

Undergraduate Researcher | shanmukhupad.github.io/portfolio/seg.pdf *May 2026 – August 2026*

- Developed a label-free PyTorch tool that maps where a DeepLabV3-ResNet50 model is likely wrong across full-size aerial images, using only the model's own confidence
- Discovered the model is about half as accurate along object edges by validating the error maps on 4 benchmark datasets, and wrote an IEEE-format paper on it

ACM SIG-CHI | github.com/sigchi-uiuc/digital-diary Urbana, IL

Project Lead *September 2025 – April 2026*

- Led a 15-person team building a Next.js journaling app that uses the Claude API, Google Calendar, and weather data
- Engineered a real-time OpenCV and MediaPipe pipeline that tracks face and hand landmarks for mood and gesture controls, deployed as its own Docker service

UIC Electronic Visualization Laboratory | github.com/ShanmukhUpad/curio Chicago, IL

Software Engineering Intern *June 2025 – August 2025*

- Cut processing time by about 30% across hundreds of Chicago urban data pipelines by building a Pandas merge tool
- Added dataset merging and multi-tab views to Curio, a TypeScript and Python web app for visual data pipelines

Projects

Multi-Region Cloud Web Service | *Node.js, Express, Python, GCP* | *Cloud Database Course Project* October 2026

- Deployed a Node.js and Express REST API on Firestore to Google Cloud servers in the US and Europe
- Quantified a 3 to 4 times slowdown in Europe by benchmarking latency in Python, since each request crossed the Atlantic twice, with 0 stale reads in 100 trials

Multi-Agent LLM Sycophancy | shanmukhupad.github.io/portfolio/mas.pdf March 2026 – May 2026

- Demonstrated that one injected false claim spreads to 100% of agents in a hierarchical LLM agentic workflow by turn 3
- Measured the failure without an LLM judge using stance flips, planning errors, and deference language, and co-authored an ACL-format paper on it

Urban AR GIS: 3D City Viewer | *Python, pyproj* | github.com/ShanmukhUpad/urban-ar-gis March 2026

- Turned OpenStreetMap data for any latitude and longitude into a 3D city by projecting building footprints to local meters with pyproj and triangulating them into 3D models
- Kept rendering at 60 FPS by running MediaPipe hand tracking on its own thread at about 30 FPS, letting users zoom, pan, and move buildings with gestures

Technical Skills

Languages: Proficient in Python, Java, C++, SQL | Familiar with C, TypeScript, JavaScript, R
Frameworks: PyTorch, Pandas, NumPy, OpenCV, React, Next.js, Node.js, Express
Infrastructure & Tools: Git, Docker, Linux, Google Cloud, Firestore, MongoDB, GeoPandas